

PBL – Jigsaw Activity: Web Server and Web Browser

Objective:

Students will develop a deep understanding of web technologies by participating in a collaborative learning activity focused on the roles and structures of web servers, web browsers, and related development tools and architectures.

Structure:

- ✓ The class is divided into 10 groups of equal size.
- ✓ Each group is assigned a specific subtopic related to web technologies.
- ✓ Students are provided with a set of guiding questions to help them explore and understand their assigned topic.
- ✓ Each group conducts a 60-minute expert discussion on their topic, during which they compile essential knowledge, examples, comparisons, and key takeaways.
- ✓ Afterward, students reform into jigsaw groups—each new group contains one member from every original group.
- ✓ Students then teach their topic to their peers and answer questions, ensuring cross-group knowledge sharing.
- ✓ The activity ends with a whole-class discussion and quizzes to reinforce learning and check understanding.

Topics and Group Assignments

Group 1: Web Server

- ✓ What is a web server?
- ✓ Examples of web servers
- ✓ Architecture of a web server
- ✓ Evolution of web server versions
- ✓ The most popular web server in 2019–2020 and comparison to others: Why did it become popular?

Group 2: Web Browser

- ✓ What is a web browser?
- ✓ Examples of web browsers
- ✓ Architecture of a web browser
- ✓ Evolution of web browser versions
- ✓ The most popular web browser in 2019–2020 and comparison to others: Why did it become popular?

Group 3: 2-Tier Architecture vs 3-Tier Architecture

- ✓ What are 2-tier and 3-tier architectures?
- ✓ Purpose of each architecture
- ✓ Explanation and comparison of both
- ✓ Advantages and disadvantages of each
- ✓ Use cases for each architecture

Group 4: GIT vs GitHub

- ✓ What is GIT?
- ✓ What is GitHub?
- ✓ Advantages and disadvantages
- ✓ Comparison: GIT vs GitHub
- ✓ How to install and configure GIT
- ✓ Basic usage and workflow

Group 5: XAMPP vs WAMP vs AMPSS Server Packages

- ✓ What are XAMPP, WAMP and AMPSS?
- ✓ Advantages and disadvantages
- ✓ Comparison: WAMP vs APPSERV
- ✓ Server package architecture
- ✓ How to install, configure, and use each

Group 6: Front-End vs Back-End Development

- ✓ Roles and responsibilities of front-end and back-end developers
- ✓ Core technologies used in each
- ✓ How front-end and back-end work together in a web application
- ✓ Front-end frameworks: React, Angular, Vue – use cases and benefits
- ✓ Back-end frameworks: Express.js, Django, Laravel – when and why to use them
- ✓ Pros and cons of using development frameworks
- ✓ Comparison table: front-end vs back-end skills, tools, and functions

Group 7: Full Stack Development

- ✓ What is a full stack developer?
- ✓ What are full stack frameworks?
- ✓ How full stack development works
- ✓ Pros and cons of full stack frameworks
- ✓ Most popular full stack frameworks

Group 8: Introduction to Containers and Docker

- ✓ What are software containers?
- ✓ Why are containers used in development and deployment?
- ✓ What is Docker?
- ✓ Key concepts: Docker images, containers, volumes, and networks
- ✓ Advantages of using Docker over traditional environments

Group 9: Docker vs Virtual Machines

- ✓ What is a virtual machine (VM)?
- ✓ Comparison between Docker containers and VMs
- ✓ Resource usage and performance differences
- ✓ Use cases for containers vs VMs
- ✓ Security considerations in both environments

Group 10: Content Management Systems (CMS)

- ✓ What is a CMS?
- ✓ Examples: WordPress, Joomla, Drupal
- ✓ How CMSs are used to build websites without coding
- ✓ Customizing themes and plugins
- ✓ Pros and cons of using a CMS